Assignment 1 Stochastic Calculus for Finance For Sunday, Feb. 15

A random walk in one dimension (with coordinate x) of length N depends on a sequence of N coin tosses. We denote the outcome of the i^{th} coin toss as ω_i with the possibilities $\omega_i = H, T$ corresponding to heads and tails.

Unless mentioned otherwise, the walker always starts at the origin $x = 0$ at time 0. The rule for the random walker is that they move $+1$ units at the i^{th} time step if $\omega_i = H$ and -1 units if $\omega_i = T$. The probability for heads is p and for tails is q with $p + q = 1$. When $p = q = \frac{1}{2}$, this is called a symmetric random walk, and when $p \neq q$, the random walk is called *asymmetric*.

This is described in terms of the random variables $\mathbb{X}_i$, $1 \leq i \leq N$, with each $\mathbb{X}_i$ being a function from the space of coin tosses $\{H, T\}$ to real numbers:

$$\mathbb{X}_i : \{H, T\} \to \mathbb{R} \ . \tag{1}$$

The random variable is defined as

$$\mathbb{X}_i(\omega_i) = \begin{cases} +1 & \text{for } \omega_i = H \ , \\ -1 & \text{for } \omega_i = T \ . \end{cases} \tag{2}$$

The location of the random walker at the n^{th} time step is also a random variable $\mathbb{M}_n$ which is simply the sum of the first n $\mathbb{X}_i$:

$$\mathbb{M}_n = \sum_{i=1}^{n} \mathbb{X}_i \ . \tag{3}$$

Since the right hand side depends on the first n coin tosses, the random variable $\mathbb{M}_n$ depends on all these n coin tosses. We shall however take a more general point of view and consider the $\mathbb{M}_n$ for all $1 \leq n \leq N$ as being functions on the sample space of all possible sequences of N coin tosses Ω_N. A typical sequence in Ω_N is denoted as $\underline{\omega} = \omega_1 \omega_2 \cdots \omega_N$. Thus, we have

$$\mathbb{M}_n : \Omega_N \to \mathbb{R} \ , \quad \text{with} \quad \mathbb{M}_n(\omega_1 \cdots \omega_N) = \sum_{i=1}^{N} \mathbb{X}_i(\omega_i) \ . \tag{4}$$

Note that even though the definition of $\mathbb{M}_n$ involves all N coin tosses, the actual values that $\mathbb{M}_n$ takes are dependent only on the first n coin tosses.

We studied the following important features of symmetric random walks in class:

1. The probability $p_{n,r}$ of a path of length n leading to (n, k) is

$$\mathbb{P}\{\mathbb{M}_n = k\} = p_{n,k} = \binom{n}{\frac{n+k}{2}} \frac{1}{2^n} \ . \tag{5}$$

2. The probability $u_{2\nu}$ of return to the origin at time 2ν:

$$\mathbb{P}\{\mathbb{M}_{2\nu} = 0\} = u_{2\nu} = \binom{2\nu}{\nu} \frac{1}{2^{2\nu}} \ . \tag{6}$$

3. The probability $f_{2\nu}$ of first return to the origin at time 2ν:

$$\mathbb{P}\{\mathbb{M}_1 \neq 0, \mathbb{M}_2 \neq 0, \ldots, \mathbb{M}_{2\nu-1} \neq 0, \mathbb{M}_{2\nu} = 0\} = f_{2\nu} = \frac{1}{2\nu - 1} \binom{2\nu}{\nu} \frac{1}{2^{2\nu}} \ . \tag{7}$$

4. The probability of the walker never returning to the origin after 2ν steps:

$$\mathbb{P}\{\mathbb{M}_1 \neq 0, \mathbb{M}_2 \neq 0, \ldots, \mathbb{M}_{2\nu-1} \neq 0, \mathbb{M}_{2\nu} \neq 0\} = u_{2\nu} = \binom{2\nu}{\nu} \frac{1}{2^{2\nu}} \ . \tag{8}$$

5. The probability $\alpha_{2\ell,2\nu}$ of last visit to the origin at time 2ℓ in a walk of length 2ν:

$$\begin{aligned} \mathbb{P}\{\mathbb{M}_{2\ell} = 0, \mathbb{M}_{2\ell+1} \neq 0, \ldots, \mathbb{M}_{2\nu} \neq 0\} = \alpha_{2\ell,2\nu} &= u_{2\ell} u_{2\nu-2\ell} \\ &= \binom{2\ell}{\ell} \frac{1}{2^{2\ell}} \binom{2(\nu-\ell)}{\nu-\ell} \frac{1}{2^{2(\nu-\ell)}} \ . \end{aligned} \tag{9}$$

Each of the above probabilities were calculated by counting the number of paths satisfying the appropriate conditions and then multiplying the number by the individual probabilities of the paths occurring. We explore more ideas related to random walks in this assignment.

1 Maxima of random walks

In this exercise, we are interested in the probability of occurrence of paths of length n which start at the origin $(0,0)$ and never cross the value $x = r \geq 0$ (but may attain $x = r$ at some point along the walk); in other words, we are interested in the probability of occurrence of paths of length n whose maximum position is at most $x = r \geq 0$. In the random variable notation, we would like to compute

$$\mathbb{P}\{\mathbb{M}_1 \leq r,\ \mathbb{M}_2 \leq r,\ \ldots,\ \mathbb{M}_n \leq r\} \ . \tag{10}$$

We split the problem into a series of steps. A typical path of this kind starts at $(0,0)$ and ends at (n,k). We need to choose $k \leq r$ since otherwise, any path will cross $x = r$ at some time and the maximum of the path will be $\geq r$. Also, recall that the maximum value k can have is $+n$ for any random path. Thus, if the length of the path $n < r$, then *every* path from $(0,0)$ to (n,k) has maximum $< r$ automatically and the required probability is 1 in this case. To get interesting answers, we need to consider path lengths n such that $n \geq r$.

So, the ranges of n and k we are interested in are

$$n \geq r \; , \quad k \leq r \; . \tag{11}$$

First, we enumerate the number of paths from $(0,0)$ to (n,k) with $k \leq r$ and such that the paths touch or cross $x = r$ at some time along the walk. This can be achieved by using a version of the reflection principle:

There is a one-to-one correspondence between paths from the origin $(0,0)$ to (n,k) which touch or cross the $x = r$ line and *all* paths from $(0,0)$ to $(n, 2r-k)$, where $(n, 2r-k)$ is the reflection of the point (n,k) across the line $x = r$.

This can be proved by first taking a path from $(0,0)$ to (n,k) which crosses or touches r and reflecting the portion of the path between the *last* crossing of r and the endpoint (n,k). This is similar to the proof of the original reflection principle discussed in class where we reflected the portion of the path between the initial point (a,α) and the *first* point where the path touches or crosses the t-axis.

Exercise 1a (2 points): Justify the above reflection principle by choosing some convenient values of r, n and k and drawing a few paths. There is no need to give a formal proof.

Exercise 1b (2 points): Using the reflection principle, show that the probability that a path of length n starts from $(0,0)$ and ends (n,k) while touching or crossing $x = r$ is

$$\mathbb{P}\{\mathbb{M}_n = 2r - k\} = p_{n,2r-k} = \binom{n}{\frac{n+2r-k}{2}} \frac{1}{2^n} \; . \tag{12}$$

As discussed earlier, the endpoint k must be $\leq r$ because we are ultimately interested in paths which never exceed r even at the endpoints.

Exercise 1c (2 points): Using the answer to the previous exercise, argue that the probability of occurrence of paths from $(0,0)$ to (n,k) with maximum exactly r is $p_{n,2r-k} - p_{n,2(r+1)-k}$.

Exercise 1d (4 points): Now we finally enumerate all possible paths with maximum $\leq r$ by summing over the allowed values of the endpoint $\mathbb{M}_n = k$. Recall that k must be $\leq r$ since we are interested only in paths which never exceed r. Also, recall that $-n \leq k \leq n$ and that we have restricted n to be $\geq r$. Thus, the upper bound for the sum over k must be r. Similarly, since the probability $p_{n,2r-k} - p_{n,2r+2-k}$ is positive only when $2r - k \leq n$, the lower bound for the sum over k is $2r - n$.

Convince yourself that the above statements are true, and show that the sum over k of the probabilities $(p_{n,2r-k} - p_{n,2r+2-k})$ evaluates to

$$\sum_{k=2r+2-n}^{r} (p_{n,2r-k} - p_{n,2r+2-k}) = p_{n,r} + p_{n,r+1} \ . \tag{13}$$

In fact, the probability of a random walk never exceeding $x = r$ in n time steps is one of $p_{n,r}$ or $p_{n,r+1}$ since either n and r have the same parity or n and $r+1$ have the same parity:

$$\mathbb{P}\{\mathbb{M}_1 \leq r,\ \mathbb{M}_2 \leq r,\ \ldots,\ \mathbb{M}_n \leq r\} = p_{n,r} \text{ or } p_{n,r+1} \ . \tag{14}$$

2 The first passage time

In a random walk, the *first passage* through the point $x = r > 0$ is said to take place at time n if the random walk satisfies

$$\mathbb{M}_1 < r \ , \quad \mathbb{M}_2 < r \ , \quad \ldots \ , \quad \mathbb{M}_{n-1} < r \ , \quad \mathbb{M}_n = r \ . \tag{15}$$

The time instant n is said to be the *first passage time*. In this exercise, we will compute the probability $\varphi_{r,n}$ for the first passage through $x = r$ to occur at time n.

Let us introduce some notation. For a particular random walk given by a sequence $\underline{\omega} = \omega_1\omega_2\cdots\omega_N$, let $\tau_r(\underline{\omega})$ be the first passage time through r for that random walk. That is, $\tau_r(\underline{\omega})$ is the time such that $\mathbb{M}_1(\underline{\omega}) < r$, $\mathbb{M}_2(\underline{\omega}) < r$, $\ldots$, $\mathbb{M}_{n-1}(\underline{\omega}) < r$ while $\mathbb{M}_{\tau_r}(\underline{\omega}) = r$. If this never happens for some given sequence $\underline{\omega}$, we define $\tau_r(\underline{\omega}) = \infty$ for that particular $\underline{\omega}$.

For this particular exercise, we shall assume that the number of coin tosses N is infinity.

Thus, τ_r is also a random variable defined on the sample space of coin tosses Ω_∞:

$$\tau_r : \Omega_\infty \to \mathbb{R} \ . \tag{16}$$

In this notation, the probability $\varphi_{r,n}$ is denoted as

$$\mathbb{P}\{\tau_r = n\} = \varphi_{r,n} \ . \tag{17}$$

That is, $\varphi_{r,n}$ is the probability that $\tau_r = n$.

Exercise 2a (2 points): A path which passes through r at time n for the first time must, in fact, have $\mathbb{M}_{n-1} = r-1$. In addition, the maximum of this path from $(0,0)$ to $(n-1, r-1)$ should be $\leq r-1$ since $\mathbb{M}_1 < r$, $\mathbb{M}_2 < r$, $\ldots$, $\mathbb{M}_{n-1} < r$.

From Exercise 1c, infer that the probability of a path from $(0,0)$ to $(n-1, r-1)$ with maximum exactly $r-1$ is $p_{n-1,r-1} - p_{n-1,r+1}$.

Exercise 2b (2 points): Using the fact that a path satisfying $\mathbb{M}_{n-1} = r-1$ can only grow to a path of length n with either $\mathbb{M}_n = r$ or $\mathbb{M}_n = r-2$, show that the probability of occurrence of a path with first passage through r at time n is

$$\mathbb{P}\{\mathbb{M}_1 < r,\ \mathbb{M}_2 < r,\ \ldots,\ \mathbb{M}_{n-1} < r,\ \mathbb{M}_n = r\} = \varphi_{r,n} = \tfrac{1}{2}(p_{n-1,r-1} - p_{n-1,r+1}) \ . \tag{18}$$

Plug in the expressions for $p_{n-1,r-1}$ and $p_{n-1,r+1}$ and obtain

$$\varphi_{r,n} = \frac{r}{n}\binom{n}{\frac{n+r}{2}}\frac{1}{2^n} \ . \tag{19}$$

Exercise 2c (4 points): Note that a path which has first passage through r at time n can be split into two sub-paths: the first sub-path starts at $(0,0)$ and has first passage through $r-1$ at some intermediate time n_1, and the second sub-path *starts* at $(n_1, r-1)$ and has a first passage through r at time n. By changing the origin of the second sub-path from $(n_1, r-1)$ to $(0,0)$ we can see that it is a typical random walk that starts at the origin $(0,0)$ and has first passage through $r = 1$ at time $n - n_1$. Thus, any path that has first passage through r at time n can be built in terms of a path that has first passage through $r-1$ at some intermediate time n_1, and a path that has first passage through 1 at time $n - n_1$.

Based on this argument, show that the probabilities $\varphi_{r,n}$ satisfy the recursion relation

$$\varphi_{r,n} = \sum_{n_1=0}^{n} \varphi_{r-1,n_1}\, \varphi_{1,n-n_1} \ . \tag{20}$$

Exercise 2d (4 points): Define the probability generating function $\Phi_r(\alpha)$ for the $\varphi_{r,n}$ as the expectation value

$$\Phi_r(\alpha) = \mathbb{E}[\alpha^{\tau_r}] = \sum_{n=0}^{\infty} \alpha^n \mathbb{P}\{\tau_r = n\} = \sum_{n=0}^{\infty} \alpha^n \varphi_{r,n} \ . \tag{21}$$

The coefficient of α^n gives the probability that the first passage time through r is given by n. Note that the coefficient of α^0 is $\varphi_{r,0}$ which we define to be 0 for all $r > 0$.

Based on the result of Exercise 2c, show that the generating functions satisfy

$$\Phi_r(\alpha) = \Phi_{r-1}(\alpha)\Phi_1(\alpha) \ , \tag{22}$$

by plugging in the series expansions nd comparing corresponding coefficients of α^n on both sides.

Applying this relation r times, we in fact get

$$\Phi_r(\alpha) = \big(\Phi_1(\alpha)\big)^r \ . \tag{23}$$

Exercise 2e (4 points): In this exercise, we evaluate the basic generating function $\Phi_1(\alpha)$. First note that $\Phi_1(\alpha)$ can be written as

$$\Phi_1(\alpha) = \sum_{n=0}^{\infty} \alpha^n \varphi_{1,n} = \alpha p + \sum_{n=2}^{\infty} \alpha^n \varphi_{1,n} \ . \tag{24}$$

Here, we have used $\varphi_{1,0} = 0$ since there is no path which starts at the origin and has first passage through 1 at time zero. Also, there is only one path with first passage through 1 at time 1 and hence its probability $\varphi_{1,1} = p$, the probability that the first coin toss results in heads.

The generating function $\Phi_1(\alpha)$ can be obtained by another judicious splitting of the random paths into three parts. Any path which has first passage through 1 at some time $n > 1$ must start with $\mathbb{M}_1 = -1$. The probability of this happening is q (taken to be $\frac{1}{2}$ for the symmetric random walk). The further journey from $(1, -1)$ to $(1, n)$ can be split into two parts. The first sub-path goes from $(1, -1)$ to $(n_1, 0)$ at some time n_1 at which it has first passage through 0, and the second sub-path starts at $(n_1, 0)$ and goes up to $(n, 1)$. Each of these two sub-paths are identical to paths starting at the origin $(0, 0)$ with first passage through 1 at times $n_1 - 1$ and $n - n_1$ respectively. Thus, we can write

$$\varphi_{1,n} = \sum_{n_1=2}^{n} q\varphi_{1,n_1-1}\varphi_{1,n-n_1} = q\varphi_{1,1}\varphi_{1,n-2} + q\varphi_{1,2}\varphi_{1,n-3} + \cdots + q\varphi_{1,n-2}\varphi_{1,1} + q\varphi_{1,n-1}\varphi_{1,0} \ . \tag{25}$$

In the last term, we have $\varphi_{1,0}$ which is zero since there can be no first passage through 1 at time zero since the walker starts at $(0, 0)$.

Note that the above relation holds for $n \geq 2$. Multiplying both sides by α^n and summing over $n = 2$ to ∞, show that we get

$$\Phi_1(\alpha) - p\alpha = q\alpha\big(\Phi_1(\alpha)\big)^2 \ . \tag{26}$$

Solve this quadratic equation for $\Phi_1(\alpha)$ to obtain

$$\Phi_1(\alpha) = \frac{1 - \sqrt{1 - 4pq\alpha^2}}{2q\alpha} \ . \tag{27}$$

The other root of the quadratic equation blows up at $\alpha = 0$ whereas we expect $\Phi_1(0) = 0$ from the definition of the generating function. This is why we chose the above root of the quadratic equation.

For the symmetric random walk, we have $p = q = \frac{1}{2}$, so we get

$$\Phi_1(\alpha) = \frac{1 - \sqrt{1 - \alpha^2}}{\alpha} \ . \tag{28}$$

From the result of Exercise 2d, we also obtain an explicit expression for the generating functions $\Phi_r(\alpha)$ for probabilities of first passage times through r:

$$\Phi_r(\alpha) = \big(\Phi_1(\alpha)\big)^r = \left(\frac{1-\sqrt{1-\alpha^2}}{\alpha}\right)^r \ . \tag{29}$$

Exercise 2f (4 points): For any given r, compute the total probability that a first passage through r occurs at *some* time or the other. That is, the probability that the first passage time through r is not infinity but some finite value $\mathbb{P}\{\tau_r < \infty\}$ (which finite value it takes is immaterial).

This requires one to compute the sum of the probabilities $\varphi_{r,n}$ over all allowed values of n for a fixed r. That is,

$$\sum_{n=r}^{\infty} \varphi_{r,n} = \sum_{n=r}^{\infty} \frac{r}{n}\binom{n}{\frac{n+r}{2}}\frac{1}{2^n} \ . \tag{30}$$

Hint: This is simply the generating function $\Phi_r(\alpha)$ evaluated at $\alpha = 1$.

What do you conclude about the probability of first passage through some r in a sufficiently long random walk?

3 The normal approximation to the binomial distribution

Recall that the probability of a symmetric random walk starting at $(0,0)$ and ending at (n,k) is $p_{n,k}$ with

$$p_{n,k} = \binom{n}{\frac{n+k}{2}}\frac{1}{2^n} \ . \tag{31}$$

The quantities n and k are related to the number of heads n_H and tails n_T in the sequence of n coin tosses as

$$n_H = \frac{n+k}{2} \ , \quad n_T = \frac{n-k}{2} \ . \tag{32}$$

So, $p_{n,k}$ is the probability that there occur n_H heads in a sequence of n coin tosses. In this context, the probabilities $p_{n,k}$ make up the *binomial distribution*. In general, the binomial distribution for n_H heads in a sequence of n coin tosses with probability p for heads and $1-p$ for tails is defined as

$$b(n_H; n, p) = \binom{n}{n_H} p^{n_H}(1-p)^{n-n_H} \ . \tag{33}$$

In this exercise, we shall get a feel for the numerical values for the $p_{n,k}$, or equivalently, the $b(n_H; n, \frac{1}{2})$ by approximating the binomial distribution by the *normal distribution*.

For a fixed number n of coin tosses and fixed probability of heads and tails, the binomial distribution is a function of the number of heads n_H. This can vary from 0 to n. For simplicity, we take the number of coin tosses to be even $n = 2\nu$, and introduce another parameter which varies symmetrically about zero:

$$r = n_H - \nu \ , \tag{34}$$

so that r varies from $-\nu$ to $+\nu$ in integer steps. We write

$$a_r = b(r + \nu; 2\nu, \tfrac{1}{2}) \ . \tag{35}$$

Exercise 3a (4 points): From the definition of $b(n_H; n, p)$, show that

$$a_r = \binom{2\nu}{\nu + r} \frac{1}{2^{2\nu}} \ , \quad \text{and establish the recursion} \quad a_r = \frac{\nu - r + 1}{\nu + r} a_{r-1} \ . \tag{36}$$

Further, show that

$$a_r = a_0 \frac{(\nu - r + 1)(\nu - r + 2) \cdots (\nu - 1)\nu}{(\nu + r)(\nu + r - 1) \cdots (\nu + 2)(\nu + 1)} \ . \tag{37}$$

Exercise 3b (4 points): In the previous formula for a_r, note that the numerator and denominator contain the same number (r) of factors. Dividing both numerator and denominator by ν^r, we get

$$a_r = a_0 \frac{(1 - \frac{r-1}{\nu})(1 - \frac{r-2}{\nu}) \cdots (1 - \frac{1}{\nu})}{(1 + \frac{r}{\nu})(1 + \frac{r-1}{\nu}) \cdots (1 + \frac{2}{\nu})(1 + \frac{1}{\nu})} \ . \tag{38}$$

Recall from the Taylor series of the exponential function e^x that one can approximate

$$\mathrm{e}^x \approx 1 + x \ , \quad \text{for small } x. \tag{39}$$

Accordingly, in the limit $\nu \to \infty$ with r fixed to be a finite integer, show that

$$a_r \approx a_0 \exp\left(-\frac{2}{\nu}[1 + 2 + \cdots + (r-1)] - \frac{r}{\nu} \right) = a_0 \, \mathrm{e}^{-r^2/\nu} \ . \tag{40}$$

Additional material: The above formula is also valid when r is not fixed in the limit $\nu \to \infty$ but is allowed to scale at most as some power of ν. The leading error in the approximation appears as a term of the form $-r^3/\nu^2$ in the exponential (not demonstrated here). So, as long as r varies in a ν-dependent range $0 < r < R(\nu)$ such that $R(\nu)^3/\nu^2 \to 0$ as $\nu \to \infty$, the above approximation for a_r will hold.

Exercise 3c (2 points): Further, a_0 itself can be approximated in the $\nu \to \infty$ limit as

$$a_0 = \binom{2\nu}{\nu} \frac{1}{2^{2\nu}} \approx \frac{1}{\sqrt{\pi \nu}} \ . \tag{41}$$

Derive this result by using Stirling's approximation for the factorial function:

$$n! \sim \sqrt{2\pi} \, n^{n+\frac{1}{2}} \mathrm{e}^{-n} \ , \tag{42}$$

where $\sim$ means that the ratio of the left hand side and right hand side tends to 1 as $n \to \infty$.

Put the results of this and the last exercise together to get

$$a_r = \binom{2\nu}{\nu + r} \frac{1}{2^{2\nu}} \approx \frac{1}{\sqrt{\pi\nu}} \mathrm{e}^{-r^2/\nu} \quad \text{in the limit } \nu \to \infty \text{ at fixed } r. \tag{43}$$

Rewriting a_r back as $b(r+\nu; 2\nu, \frac{1}{2})$ and then back in terms of $p_{2\nu,k}$, show that

$$p_{2\nu,k} \approx \frac{1}{2\sqrt{\pi\nu}} \mathrm{e}^{-k^2/4\nu} \ . \tag{44}$$

(Since $r = k/2$, the density function for k will in fact be 1/2 that of r,with both treated approximately as continuous variables.)

This result shows that the location k of the walker at the end of $n = 2\nu$ times steps for large n is normally distributed with average 0 and standard deviation $\sqrt{n}$. In the limit of large ν, the standard deviation also becomes very large. This indicates that the location of the random walker has increasingly more spread "on average" ($\approx \sqrt{n}$) as one goes to longer and longer times. In the continuum limit where time becomes a continuous time parameter, this fact implies a typical random path is not differentiable in the usual sense and hence requires the introduction of stochastic calculus to make sense of time derivatives.

□